ZZ-B-616D
August 23, 1976
SUPERSEDING
Int.Fed.Spec. ZZ-B-00616c(DSA-DM)
December 9, 1969 and
Fed. Spec. ZZ-B-616b
May 14, 1957

FEDERAL SPECIFICATION

BOWL, PLASTER OF PARIS MIXING

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers bowls suitable for mixing dental gypsum products and alginate hydrocolloids with water.

1.2 Classification. Bowls shall be furnished in the following sizes, as specified (see 6.1):

Extra large
Large
Medium
Small

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.

FSC 6520

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5.7 This specification covers the following

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications, Standards and Publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Standards (ASTM):

D412 - Tension Testing of Vulcanized Rubber.
D573 - Test for Accelerated Aging of Vulcanized Rubber by the Oven Method.

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D792 - Tests for Specific Gravity and Density of Plastics by Displacement.
D2240 - Test for Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Material. Bowls shall be made of flexible, synthetic vinyl elastomeric material with surface release characteristics which prevent customary materials, such as dental plaster, stone, and alginate when mixed in the plaster bowl, from adhering to its surface, even when the materials have hardened and set in the bowl.

3.2 Physical properties. The vinyl elastomeric material shall conform to the physical properties listed in table I when tested as specified in 4.4.1.

TABLE I. Physical properties

Characteristic	Value
Hardness (Durometer Shore A)	54 to 78
Specific Gravity	1.21 to 1.29
Tensile strength, pounds per square inch, (p.s.i.) minimum	1640 p.s.i.
Ultimate elongation (percent (%) minimum)	380%
Elongation after air-over aging (% minimum)	310%

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be zero for all sample sizes and a minimum

3.3 Description. Bowls shall be of such design, strength, and flexibility as to provide a sturdy container for vigorous spatulations of alginates and gypsum materials and upon flexing the bowl it shall form a pouring lip to expel the fluid material. Bowls shall be furnished in four sizes: extra large, large, medium, and small (see 1.2 and 6.1). Bowls shall conform to the shape shown in figure 1 and the dimensions listed in table II for the applicable size specified.

3.3.1 Unit of issue. Unless otherwise specified (see 6.1), one (each) mixing bowl of the size specified, constitutes one unit of issue.

3.4 Color. Color of bowls shall be at the option of the contractor, provided each bowl is one uniform, solid color.

3.5 Surface adherences. There shall be no adherence of mixed gypsum or alginate material to the surface of the mixing bowl, except for discrete particles which may be removed with a moistened finger, when tested as specified in 4.4.2.

3.6 Capacity. When filled to 1/4 inch from the top, the bowls shall hold the following minimum amounts of water for the respective size when tested as specified in 4.4.3.

Extra large	- 1550 milliliters
Large	- 525 milliliters
Medium	- 300 milliliters
Small	- 100 milliliters

3.7 Finish. Bowls shall have a smooth, glossy surface, inside and outside, except for the outside of the base which is used for identification marking.

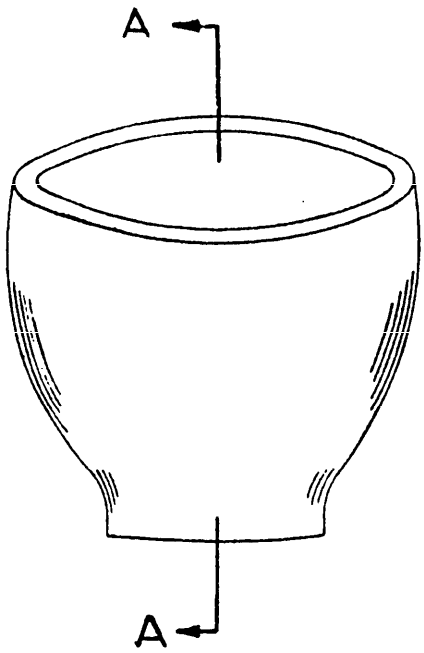
3.8 Identification marking. Bowls shall be permanently and legibly molded on the outside of the base with the name or trademark of the manufacturer or contractor. The outside of the base of the bowl shall be marked with the manufacturer's or contractor's size designation.

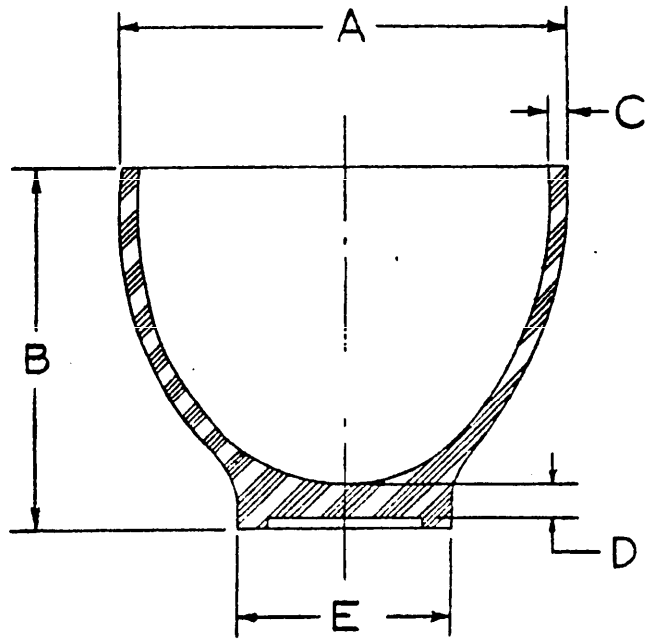
3.9 Workmanship. Workmanship shall be first class throughout. Bowls shall be free from blisters, cracks, embedded foreign matter, or other defects which may detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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SECTION A-A

TABLE II DIMENSIONS (INCHES)

SPECIFICATION DESIGNATION SIZE	OUTSIDE DIA AT TOP A	OVERALL HEIGHT B	MIN THK OF SIDE C	MIN THK OF BASE D 1/	OUTSIDE DIA OF BASE E
EXTRA LARGE	$6\frac{3}{8} \pm \frac{1}{4}$	$5\frac{3}{4} \pm \frac{1}{4}$	$\frac{3}{16}$	$\frac{1}{8}$	$2\frac{3}{8} \pm \frac{1}{8}$
LARGE	$4\frac{3}{4} \pm \frac{1}{4}$	$3\frac{3}{4} \pm \frac{1}{4}$	$\frac{9}{64}$	$\frac{1}{8}$	$2\frac{1}{4} \pm \frac{1}{8}$
MEDIUM	$4 +\frac{3}{8}, -\frac{1}{4}$	$2\frac{7}{8} \pm \frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$2 +\frac{9}{64}, -\frac{1}{8}$
SMALL	$3\frac{1}{4} \pm \frac{1}{4}$	$2\frac{1}{8} \pm \frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$1\frac{3}{4} \pm \frac{1}{8}$

1/ DIMENSION SHALL BE MEASURED IN BETWEEN ANY IDENTIFICATION MARKING WHICH MAY BE PRESENT ON OUTSIDE OF BASE.

BOWL, PLASTER OF PARIS MIXING

FIGURE 1

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the vinyl elastomeric materials for the physical properties specified in table I, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II, IV and V (see 4.2.1, 4.3, and 4.4.1) and table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. Unit of product for examination purposes shall be one bowl.

TABLE II. Inspection and acceptable quality levels

Type defect	Inspection level	AQL (percent defective)
Major	S-2	2.5
Minor A	S-2	4.0
Minor B	S-1	4.0

4.2.2 Sampling for tests on raw materials. In the absence of certificates of quality, sampling for tests specified in 4.4.1 shall be conducted in accordance with table III. The sample unit shall be as specified in the applicable American Society for Testing and Materials (ASTM) test method. Samples shall be taken from each lot of vinyl elastomeric material used in the construction of the bowls, and results shall be reported in accordance with the applicable ASTM method.

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TABLE III. Tests

Characteristic	Requirements	Test procedure
Hardness	3.2	4.4.1
Specific gravity	3.2	4.4.1
Tensile strength	3.2	4.4.1
Ultimate elongation	3.2	4.4.1
Elongation after air-oven aging	3.2	4.4.1

4.2.3 Sampling for surface adherence test. Sampling for capacity testing and surface adherence test shall be conducted on the end item in accordance with MIL-STD-105, inspection level S-1, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one bowl.

4.3 Examination. Bowls shall be examined for defects including, but not limited to, those listed in table IV.

TABLE IV. Classification of defects

Categories	Defects
Major:	
101	Shape not as shown on figure.
102	Side of bowls cannot be formed into a pouring lip.
103	Tears, cracks, blisters, or embedded foreign matter present in bowls.
104	Surface finish not as specified.
Minor A:	
201	Identification marking not present or properly located; not legible and permanent.
202	Identification marking not correct for applicable size bowl.
203	Color not one solid color.
Minor B:	
251	Dimensions not as specified.

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4.3.1 Packaging inspection. The inspection of the preservation - packaging and interior package marking shall be in accordance with group A quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests.

4.4.1 Tests for vinyl elastomeric material. Vinyl elastomeric material shall be tested for the following physical characteristics in accordance with the applicable American Society for Testing and Materials (ASTM) Test Method as specified in table V, except that air-oven aging test shall be conducted for a period of 7 days at a temperature of 100 degrees Centigrade.

TABLE V. Test methods

Characteristic	ASTM Test Method
Hardness	ASTM D2240
Specific gravity	ASTM D792
Tensile strength	ASTM D412
Ultimate elongation	ASTM D412
Elongation after air-oven aging	ASTM D573

4.4.2 Surface adherence. A mix of plaster or alginate material with water shall be made according to manufacturer's instructions in the mixing bowl. The material shall be allowed to set and then shall be removed by squeezing the sides of the bowl.

4.4.3 Capacity testing. Bowls shall be filled with water to a level 1/4 inch from the top of the bowl. A graduated cylinder shall be used to transfer the water and to determine the amount of water (in milliliters) needed to fill the bowl to a level 1/4 inch from the top.

5. PREPARATION FOR DELIVERY

5.1 Preservation - packaging. Preservation - packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each bowl shall be packaged (method III of MIL-P-116) in a box of appropriate size and constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

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5.1.1.2 Intermediate packaging. The number of bowls specified in table VI of the size specified shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Bowls shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

TABLE VI. Packaging and packing quantities

Size	Intermediate package	Exterior container
Extra large	12	24
Large	12	144
Medium	12	288
Small	12	360

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. The number of bowls specified in table VI of the size specified shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, type CF, class weather resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with PPP-B-636.

5.2.2 Level B. The number of bowls specified in table VI of the size specified shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. Bowls in the quantity as specified shall be packed in a container complying with the Uniform Freight Classification or National Motor Freight Classification, as applicable. The container shall be packed to assure carrier acceptance and safe arrival at destination.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

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5.3.2 Military activities. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) If unit of issue other than that specified (see 3.3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Activity:
VA-DMS
GSA - FSS

Preparing activity:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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